Develop a MATLAB model of a microwave filter and analyze the effect of non-ideal filter characteristics on signal transmission.

MATLAB Code:

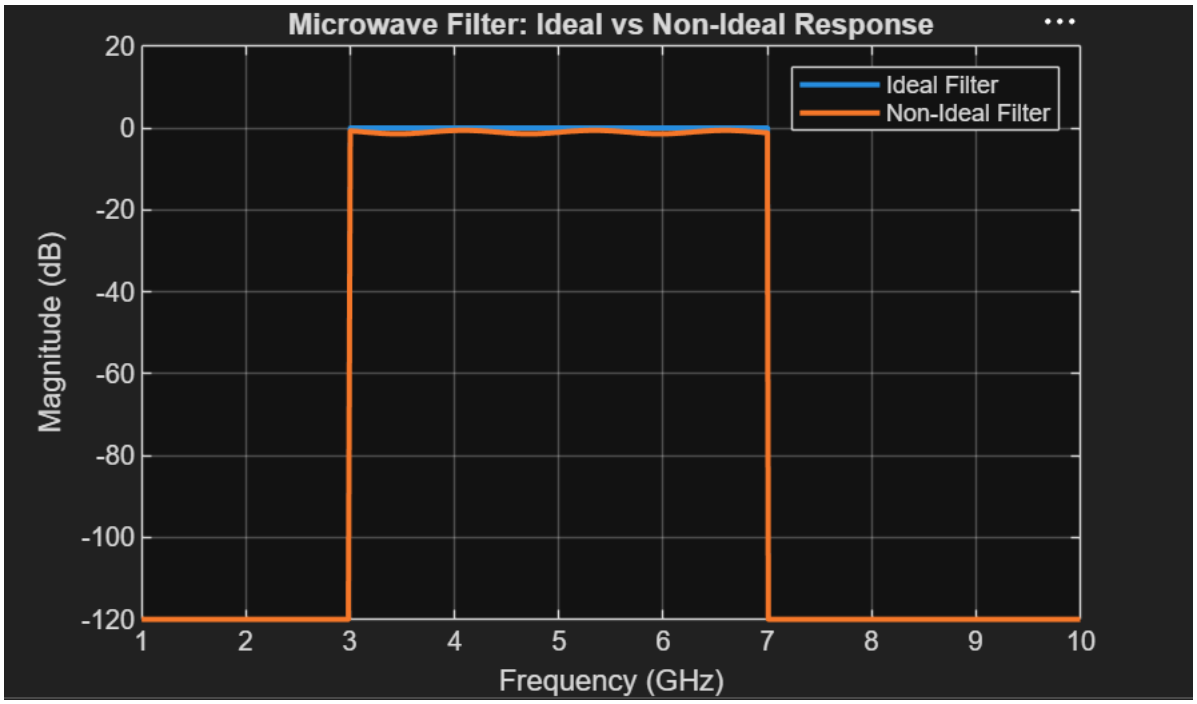
```
clc;
clear;
close all;
% Frequency range
f = linspace(1, 10, 1000);   % GHz
% Ideal band-pass filter
fc1 = 3;                  % Lower cutoff frequency
fc2 = 7;                  % Upper cutoff frequency
H_ideal = zeros(size(f));
H_ideal(f >= fc1 & f <= fc2) = 1;
% Non-ideal filter
ripple = 0.5;               % Passband ripple in dB
loss = 1;                 % Insertion loss in dB
H_nonideal = H_ideal .* ...
    10.^(-loss/20) .* ...
    (1 + 0.05*sin(5*f));
% Convert to dB
H_ideal_dB = 20*log10(H_ideal + 1e-6);
H_nonideal_dB = 20*log10(abs(H_nonideal) + 1e-6);
% Plot response
figure;
plot(f,H_ideal_dB,'LineWidth',2);
hold on;
plot(f,H_nonideal_dB,'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
```

ylabel('Magnitude (dB)');

title('Microwave Filter: Ideal vs Non-Ideal Response');

legend('Ideal Filter','Non-Ideal Filter');

MATLAB OUTPUT:



RESULT:

The graph shows that:

- The ideal filter has a flat response between 3–7 GHz.
- The non-ideal filter has reduced signal strength due to insertion loss.
- Small variations occur in the passband because of ripple.
- Therefore, non-ideal characteristics reduce the quality of signal transmission.